

Assignment 1: Vision Transformers

Implementation and Analysis Report

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Varun Teja Goud Madhagouni

varuntej@buffalo.edu

#50672516

1 Overview

This report documents the implementation of a Vision Transformer (ViT) trained on a 5,000-image class-balanced subset of CIFAR-10, followed by ablation studies and quantitative analysis. The baseline model uses patch size 4, embedding dimension 64, 2 attention heads, 4 Transformer layers, MLP hidden dimension 128, dropout 0.0, batch size 64, learning rate 10^{-3} , and trains for 20 epochs with AdamW and cosine annealing, yielding 142,026 trainable parameters and a final validation accuracy of **47.72%**.

2 Section A: Ablation Analysis

2.1 A.1 Patch Size Ablation

I trained three models with patch sizes $P \in \{2, 4, 8\}$, having all other hyperparameters fixed. Table 1 summarizes the results.

Table 1: Patch size ablation: final accuracy and parameter count.

P	Tokens N	Params	Val Acc
2	256	152,010	48.50%
4	64	142,026	47.72%
8	16	148,170	40.25%

Patch size $P = 2$ achieves the highest accuracy at 48.50%, compared to 47.72% for $P = 4$ and only 40.25% for $P = 8$. The gain from $P = 4$ to $P = 2$ is marginal, while the drop from $P = 4$ to $P = 8$ is huge. The parameter counts are non-monotone across patch sizes because the patch embedding layer has $P^2 \cdot C \cdot D$ parameters, which grows with P , while the positional embedding has $(N + 1) \cdot D$ parameters, which shrinks with P .

The core trade-off is between *token resolution* and *sequence length*. A smaller patch size produces more tokens ($N = 256$ vs. $N = 64$ vs. $N = 16$), each covering a finer spatial region. However, more tokens also means quadratic increase in attention computations ($O(N^2)$) and longer sequences that are harder to optimize within 20 epochs on only a 5,000 training image dataset. For $P = 8$, each token summarizes an 8×8 pixel region over 1/4th of the 32×32 image therefore many features are kept in a single path making attention mechanism less significant. The $P = 2$ advantage is modest because the model is data-limited: validation accuracy for $P = 2$ plateaued around epoch 13 (48.35%) and barely improved to 48.50% by epoch 20, while training loss fell to 0.68, maybe leading to over-fitting.

2.2 A.2 Number of Layers Ablation

I trained three models with $L \in \{2, 4, 8\}$ Transformer layers. Table 2 shows the results.

Table 2: Layer depth ablation results.

Layers L	Params	Val Acc
2	75,082	46.95%
4	142,026	47.72%
8	275,914	46.12%

Accuracy does *not* improve linearly with depth. The 4-layer model achieves 47.72%, outperformed both the 2-layer model (46.95%) and the 8-layer model (46.12%). The ordering is $L=4 > L=2 > L=8$, with the 8-layer model performing worst despite having nearly $4\times$ more parameters than the 2-layer model.

Going from 2 to 4 layers yields a +0.77 pp gain, as the additional sub-layers allow the model to compose more abstract features. However, going from 4 to 8 layers hurts accuracy by -1.60 pp despite lower training loss at epoch 20 (0.634 for $L=8$ vs. 0.614 for $L=4$). This is

a classic example of overfitting: the 8-layer model has 275,914 parameters but is trained on only 5,000 examples. Its validation accuracy peaked at 45.69% at epoch 11 and then oscillated between 43.8% and 46.0% through epoch 20, a classic overfitting trajectory. Without regularization (dropout is 0.0 in all runs), its maybe because deeper models memorize the small training set rather than generalizing. These results suggest that 4 layers is the sweet spot between capacity and generalization at this dataset scale.

2.3 A.3 Positional Encoding Ablation

Removing positional embeddings reduces final validation accuracy from 47.72% (baseline) to **43.89%**, a drop of 3.83 percentage points. Table 3 compares training trajectories at key epochs.

Table 3: Baseline vs. no positional encoding: validation accuracy.

Epoch	Baseline	No Pos. Enc.
1	27.30%	26.91%
5	40.74%	39.81%
10	45.57%	41.13%
15	47.06%	43.87%
20	47.72%	43.89%

Both models start at a similar level but diverge after epoch 5. The baseline consistently improves through epoch 20, while the no-positional-encoding model staginates from epoch 11 onward (accuracy oscillates between 43.70% and 43.91% for epochs 11–20).

The Transformer self-attention mechanism is *permutation-invariant*: without positional embeddings, every permutation of the 64 patches produces exactly the same CLS representation, so the model cannot distinguish spatial arrangement. This forces the model to rely entirely on *bag-of-patches* statistics. For CIFAR-10, where object shape and layout carry significant discriminative information, this is a meaningful handicap. The 3.83 pp drop confirms that spatial structure is important even at 32×32 resolution.

3 Section B: Quantitative Analysis

3.1 B.1 Attention Entropy

Table 4 reports the Shannon entropy of the CLS token’s attention distribution for each layer, averaged over all heads and all 10,000 test images.

Table 4: CLS attention entropy (bits) per layer.

Layer	Entropy (bits)
0	5.8576
1	5.9234
2	5.9343
3	5.9989

Entropy *increases* linearly with depth, rising from 5.8576 bits at layer 0 to 5.9989 bits at layer 3 a 2.41% increase. This is counterintuitive relative to the expectation that deeper layers focus attention more sharply. The maximum possible entropy for a uniform distribution over $T = 65$ tokens is $\log_2(65) \approx 6.022$ bits, so all layers are operating near the maximum, the CLS token attends nearly uniformly over all tokens in every layer.

This solidifies that attention propagates more strongly as the number of layers increase.

3.2 B.2 Positional Embedding Correlation

The Pearson correlation between pairwise cosine similarities of patch positional embeddings and pairwise Euclidean grid distances is:

$$r = -0.3408, \quad n_{\text{pairs}} = 2,016$$

The correlation is **negative**, as expected. A negative r means that spatially close patches (small Euclidean distance) tend to have more similar positional embeddings (high cosine similarity), and distant patches tend to have dissimilar embeddings, the desired spatial coherence property.

The magnitude $|r| = 0.3408$ indicates a moderate but statistically significant correlation (with 2,016 pairs). This moderate value is sensible: zero-initialized positional embeddings trained from scratch on 5,000 images for 20 epochs have only partially developed smooth 2-D spatial structure. Nonetheless, the negative sign confirms that the model has learned that patch k at grid position (r_k, c_k) should have a positional representation similar to its spatial neighbors.

3.3 B.3 Per-Class Accuracy and Confusions

Table 5 shows per-class test accuracy. Table 6 shows the top-3 misclassification pairs.

Cat (27.60%) and Bird (36.10%) are the hardest classes; Ship (63.60%) is the easiest. The #1 confusion 232 cats misclassified as dogs, is the largest single off-diagonal

Table 5: Per-class test accuracy on the full 10,000-image test set.

Label	Class	Accuracy
0	Airplane	54.50%
1	Automobile	56.90%
2	Bird	36.10%
3	Cat	27.60%
4	Deer	36.40%
5	Dog	39.90%
6	Frog	56.40%
7	Horse	50.30%
8	Ship	63.60%
9	Truck	55.50%
Mean		47.72%

Table 6: Top-3 misclassification pairs (true $\rightarrow$ predicted, count).

True	Predicted	Count
Cat (3)	Dog (5)	232
Truck (9)	Automobile (1)	200
Auto (1)	Truck (9)	194

entry in the confusion matrix, reflecting the well-known visual similarity between the two classes at low resolution. The #2 and #3 confusions form a symmetric Truck $\leftrightarrow$ Automobile pair (200 and 194 errors respectively), expected since both are motor vehicles with similar looks at 32×32 . Ship’s high accuracy is explained by its distinctive visual context: water backgrounds provide a strong separating.

3.4 B.4 Mean Attention Distance

Table 7 reports the mean attention distance (in patch units) for each layer.

Table 7: Mean attention distance per layer (patch units, $G = 8$).

Layer	Mean Distance
0	4.1281
1	4.1147
2	4.0994
3	4.1760

The mean attention distances are remarkably stable across all layers, spanning only 4.0994 to 4.1760, a total spread of 0.077 patch units. The maximum possible distance on an 8×8 grid is $\sqrt{7^2 + 7^2} \approx 9.9$ patches, and the expected distance under a uniform distribution is

approximately 3.6 to 4.2 patches, so all layers are consistent with near-uniform global attention. This aligns with the high entropy values in B.1.

This contrasts sharply with convolutional networks, where the receptive field grows strictly with depth: a 3×3 conv at layer l sees a receptive field of $2l+1$ pixels, growing by exactly 2 pixels per layer. In a ViT, all 64 tokens are mutually visible from layer 0, so there is no structural forcing function to grow the receptive field. Large pretrained ViTs develop local heads in early layers and global heads in later layers, creating a coarse-to-fine hierarchy. Our model has not developed this pattern: all four layers attend globally and uniformly, treating attention as all-to-all pooling rather than a spatially structured extractor, a behavior attributable to limited training data and the absence of pretraining.

4 Conclusion

A compact ViT (142K parameters) trained from scratch on 5,000 CIFAR-10 images achieves 47.72% accuracy. Among the three ablations, patch size has the largest impact (40.25%–48.50%), followed by positional encoding removal (-3.83 pp), then depth (-1.60 pp for over-deep models). The quantitative analysis reveals that this small ViT operates in a near-uniform attention regime across all layers. entropy reaches 97–99.5% of the maximum and mean attention distances are nearly constant indicating that the hierarchical, spatially specialized attention patterns seen in large pretrained ViTs require substantially more data or pretraining to emerge.